

# KIT SAFETY DATA SHEET



**Kit Product Name** Autoimmune EIA Anti-SS-A/Ro Test

**Kit Catalogue Number(s)** 96SA

**Revision date** 01-Dec-2021

## Kit Contents

| Catalogue Number(s)            | Product Name                                |
|--------------------------------|---------------------------------------------|
| 200SA, 210SA                   | Anti-SS-A/Ro Pos. Cntrl/Anti-SS-A/Ro Calib. |
| 220NC, 220ND                   | Negative Control                            |
| 220HSP, 220HAN, 220HDS, 220HCE | Conjugate                                   |
| 230AW                          | Wash Concentrate                            |
| 230AD                          | Sample Diluent                              |
| 220TM                          | Substrate                                   |
| 220SM                          | Stop Solution                               |



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
The Globally Harmonised System of Classification and Labelling of Chemicals (GHS)

Revision date 02-Sep-2021

Revision Number 1

## 1. IDENTIFICATION

### Product identifier

**Product Name** Anti-SS-A/Ro Pos. Cntrl/Anti-SS-A/Ro Calib.

### Other means of identification

**Catalogue Number(s)** 200SA, 210SA

**Registration Number(s)** No information available

### Recommended use of the chemical and restrictions on use

**Recommended use** In-vitro laboratory reagent or component

### Supplier's details

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Diagnostic Group  
4000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pvt. Ltd.  
Bio-Rad House  
86-87, Udyog Vihar Phase IV Gurgaon  
122005  
Haryana India

Bio-Rad Laboratories (Pty) Ltd.  
34 Bolton Road  
Parkwood, Johannesburg 2193  
South Africa

### Technical Service

India: 91-124-4029300 or 1-800-180-1224  
South Africa: 27-11-442-85-08  
India: support.india@bio-rad.com  
South Africa: cdg\_techsupport\_eemea@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC India: 000-800-100-7141  
CHEMTREC South Africa: 0-800-983-611

## 2. HAZARDS IDENTIFICATION

### Classification of the substance or mixture

Acute aquatic toxicity

Category 3

### GHS Label elements, including precautionary statements

#### **Hazard statements**

Harmful to aquatic life

#### **Precautionary Statements - Prevention**

Avoid release to the environment

#### **Precautionary Statements - Response**

#### **Precautionary Statements - Disposal**

Dispose of contents/container to an approved waste disposal plant

### Other hazards which do not result in classification

No information available

Contains human source material and / or potentially infectious components

### 3. COMPOSITION/INFORMATION ON INGREDIENTS

**Substance**

Not applicable

**Mixture**

| Chemical name                 | CAS No  | Weight-% |
|-------------------------------|---------|----------|
| 1,2,3-Propanetriol<br>56-81-5 | 56-81-5 | 52.717   |

### 4. FIRST AID MEASURES

**Description of necessary first aid measures**

|              |                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                 |
| Skin contact | Wash skin with soap and water. In the case of skin irritation or allergic reactions see a physician.                 |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| Ingestion    | Call a physician.                                                                                                    |

**For emergency responders**

|                                    |                           |
|------------------------------------|---------------------------|
| Self-protection of the first aider | No information available. |
|------------------------------------|---------------------------|

**Most important symptoms and effects, both acute and delayed**

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

**Indication of immediate medical attention and special treatment needed, if necessary**

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. FIREFIGHTING MEASURES

**Suitable Extinguishing Media**

|                              |                                                                                                         |
|------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|------------------------------|---------------------------------------------------------------------------------------------------------|

|            |                                                                    |
|------------|--------------------------------------------------------------------|
| Large Fire | CAUTION: Use of water spray when fighting fire may be inefficient. |
|------------|--------------------------------------------------------------------|

|                                |                                                                   |
|--------------------------------|-------------------------------------------------------------------|
| Unsuitable extinguishing media | Do not scatter spilled material with high pressure water streams. |
|--------------------------------|-------------------------------------------------------------------|

**Specific hazards arising from the chemical**

|                                            |                           |
|--------------------------------------------|---------------------------|
| Specific hazards arising from the chemical | No information available. |
|--------------------------------------------|---------------------------|

**Special protective actions for fire-fighters**

|                                                                |                                                                                                                                    |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|

### 6. ACCIDENTAL RELEASE MEASURES

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Ensure adequate ventilation.

**Environmental precautions**

**Environmental precautions** See Section 12 for additional Ecological Information.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Clean contaminated surface thoroughly.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

**7. HANDLING AND STORAGE****Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place.

**Incompatible materials** Metals.

**8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control parameters**

Exposure guidelines

| Chemical name                 | ACGIH TLV         | OSHA PEL                                                                                                                                                                                                                               | Ontario                   | European Union       |        |
|-------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------|--------|
| 1,2,3-Propanetriol<br>56-81-5 | No data available | TWA: 15 mg/m <sup>3</sup> mist, total particulate<br>TWA: 5 mg/m <sup>3</sup> mist, respirable fraction<br>(vacated) TWA: 10 mg/m <sup>3</sup> mist, total particulate<br>(vacated) TWA: 5 mg/m <sup>3</sup> mist, respirable fraction | -                         | -                    |        |
| Chemical name                 | China             | Japan Society of Occupational Health                                                                                                                                                                                                   | OEL                       | Australia            | Taiwan |
| 1,2,3-Propanetriol<br>56-81-5 | -                 | -                                                                                                                                                                                                                                      | TWA: 10 mg/m <sup>3</sup> | 10 mg/m <sup>3</sup> | -      |

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Skin and body protection** Wear suitable protective clothing.

**Hand protection** Wear suitable gloves.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

## 9. PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                              |                   |                          |                          |
|----------------------------------------------|-------------------|--------------------------|--------------------------|
| Physical state                               | Liquid            | Odour                    | Odourless                |
| Appearance                                   | aqueous solution  | Odour threshold          | No information available |
| Colour                                       | white             |                          |                          |
| <u>Property</u>                              | <u>Values</u>     | <u>Remarks • Method</u>  |                          |
| pH                                           |                   | No information available |                          |
| Melting point / freezing point               |                   | No information available |                          |
| Boiling point / boiling range                | > 100 °C / 212 °F |                          |                          |
| Flash point                                  | > 160 °C / 320 °F |                          |                          |
| Evaporation rate                             |                   | No information available |                          |
| Flammability (solid, gas)                    |                   | No information available |                          |
| Upper/lower flammability or explosive limits |                   |                          |                          |
| Upper flammability or explosive limits       | Not applicable    |                          |                          |
| Lower flammability or explosive limits       | Not applicable    |                          |                          |
| Vapour pressure                              |                   | No information available |                          |
| Vapour density                               |                   | No information available |                          |
| Relative density                             |                   | No information available |                          |
| Solubility(ies)                              |                   |                          |                          |
| Water solubility                             | Miscible in water |                          |                          |
| Solubility in other solvents                 |                   | No information available |                          |
| Partition coefficient                        |                   | No information available |                          |
| Autoignition temperature                     |                   | No information available |                          |
| Decomposition temperature                    |                   | No information available |                          |
| Viscosity                                    |                   |                          |                          |
| Kinematic viscosity                          |                   | No information available |                          |
| Dynamic viscosity                            |                   |                          |                          |

### Other information

Oxidising properties                      Not applicable

## 10. STABILITY AND REACTIVITY

### Reactivity

Reactivity                                      No information available.

### Chemical stability

Stability                                        Stable under normal conditions.

### Explosion data

Sensitivity to mechanical impact    None  
Sensitivity to static discharge        None.

### Possibility of hazardous reactions

Possibility of hazardous reactions    None under normal processing.

### Conditions to avoid

Conditions to avoid                        None known based on information supplied.

### Incompatible materials

Incompatible materials                    Metals.

**Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

## 11. TOXICOLOGICAL INFORMATION

**Information on the likely routes of exposure****Product Information**

|                     |                                                                   |
|---------------------|-------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available. |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available. |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. |

**Symptoms** No information available.

**Acute toxicity****Numerical measures of toxicity**

0.001 % of the mixture consists of ingredient(s) of unknown acute oral toxicity  
 0.001 % of the mixture consists of ingredient(s) of unknown acute dermal toxicity  
 52.718 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (gas)  
 52.718 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (vapor)  
 0.001 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (dust/mist)

**Component Information**

| Chemical name      | Oral LD50             | Dermal LD50          | Inhalation LC50                     |
|--------------------|-----------------------|----------------------|-------------------------------------|
| 1,2,3-Propanetriol | = 12600 mg/kg ( Rat ) | > 10 g/kg ( Rabbit ) | > 570 mg/m <sup>3</sup> ( Rat ) 1 h |

**Delayed and immediate effects and also chronic effects from short and long term exposure**

|                                          |                                         |
|------------------------------------------|-----------------------------------------|
| <b>Skin corrosion/irritation</b>         | No information available.               |
| <b>Serious eye damage/irritation</b>     | No information available.               |
| <b>Respiratory or skin sensitization</b> | No information available.               |
| <b>Germ cell mutagenicity</b>            | No information available.               |
| <b>Carcinogenicity</b>                   | No information available.               |
| <b>Reproductive toxicity</b>             | No information available.               |
| <b>STOT - single exposure</b>            | No information available.               |
| <b>STOT - repeated exposure</b>          | No information available.               |
| <b>Target organ effects</b>              | Kidney, Respiratory system, Eyes, Skin. |
| <b>Aspiration hazard</b>                 | No information available.               |

## 12. ECOLOGICAL INFORMATION

**Toxicity**

0.001 % of the mixture consists of component(s) of unknown hazards to the aquatic environment

**Ecotoxicity** Harmful to aquatic life.

| Chemical name      | Algae/aquatic plants | Fish                                                 | Crustacea                                   |
|--------------------|----------------------|------------------------------------------------------|---------------------------------------------|
| 1,2,3-Propanetriol | -                    | LC50: 51 - 57mL/L (96h, <i>Oncorhynchus mykiss</i> ) | EC50: >500mg/L (24h, <i>Daphnia magna</i> ) |

#### Persistence and degradability

No information available.

#### Bioaccumulative potential

There is no data for this product.

#### Mobility

**Mobility in soil** No information available.

**Mobility** No information available.

| Chemical name      | Partition coefficient |
|--------------------|-----------------------|
| 1,2,3-Propanetriol | -1.76                 |

#### Other adverse effects

No information available.

### 13. DISPOSAL CONSIDERATIONS

#### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

### 14. TRANSPORT INFORMATION

**IMDG** Not regulated  
**Transport in bulk according to Annex II of MARPOL and the IBC Code** No information available

**IATA** Not regulated

**RID** Not regulated

**ADR** Not regulated

**ADN** Not regulated

**Special precautions for user** Special provisions from the regulations relative to the specified mode of transport are noted by numeric code. Refer to the regulations for the full text of special provisions.

### 15. REGULATORY INFORMATION

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

**International Inventories**

Contact supplier for inventory compliance status

## 16. OTHER INFORMATION

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 02-Sep-2021

**Revision Note** SDS sections updated. 1. 2.

**Key or legend to abbreviations and acronyms used in the safety data sheet**

**Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**





# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
The Globally Harmonised System of Classification and Labelling of Chemicals (GHS)

Revision date 01-Sep-2021

Revision Number 1

## 1. IDENTIFICATION

### Product identifier

**Product Name** Negative Control

### Other means of identification

**Catalogue Number(s)** 220NC, 220ND

**Registration Number(s)** No information available

### Recommended use of the chemical and restrictions on use

**Recommended use** In-vitro laboratory reagent or component

### Supplier's details

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Diagnostic Group  
4000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pvt. Ltd.  
Bio-Rad House  
86-87, Udyog Vihar Phase IV Gurgaon  
122005  
Haryana India

Bio-Rad Laboratories (Pty) Ltd.  
34 Bolton Road  
Parkwood, Johannesburg 2193  
South Africa

### Technical Service

India: 91-124-4029300 or 1-800-180-1224  
South Africa: 27-11-442-85-08  
India: support.india@bio-rad.com  
South Africa: cdg\_techsupport\_eemea@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC India: 000-800-100-7141  
CHEMTREC South Africa: 0-800-983-611

## 2. HAZARDS IDENTIFICATION

### Classification of the substance or mixture

|                        |            |
|------------------------|------------|
| Acute aquatic toxicity | Category 3 |
|------------------------|------------|

### GHS Label elements, including precautionary statements

#### **Hazard statements**

Harmful to aquatic life

#### **Precautionary Statements - Prevention**

Avoid release to the environment

#### **Precautionary Statements - Response**

#### **Precautionary Statements - Disposal**

Dispose of contents/container to an approved waste disposal plant

### Other hazards which do not result in classification

No information available

Contains human source material and / or potentially infectious components

### 3. COMPOSITION/INFORMATION ON INGREDIENTS

#### Substance

Not applicable

#### Mixture

| Chemical name                 | CAS No  | Weight-% |
|-------------------------------|---------|----------|
| 1,2,3-Propanetriol<br>56-81-5 | 56-81-5 | 29.467   |

### 4. FIRST AID MEASURES

#### Description of necessary first aid measures

|              |                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                 |
| Skin contact | Wash skin with soap and water. In the case of skin irritation or allergic reactions see a physician.                 |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| Ingestion    | Call a physician.                                                                                                    |

#### For emergency responders

Self-protection of the first aider No information available.

#### Most important symptoms and effects, both acute and delayed

Symptoms No information available.

#### Indication of immediate medical attention and special treatment needed, if necessary

Note to physicians Treat symptomatically.

### 5. FIREFIGHTING MEASURES

#### Suitable Extinguishing Media

Suitable Extinguishing Media Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Large Fire CAUTION: Use of water spray when fighting fire may be inefficient.

Unsuitable extinguishing media Do not scatter spilled material with high pressure water streams.

#### Specific hazards arising from the chemical

Specific hazards arising from the chemical No information available.

#### Special protective actions for fire-fighters

Special protective equipment and precautions for fire-fighters Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

### 6. ACCIDENTAL RELEASE MEASURES

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Ensure adequate ventilation.

**Environmental precautions**

**Environmental precautions** See Section 12 for additional Ecological Information.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Clean contaminated surface thoroughly.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## 7. HANDLING AND STORAGE

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place.

## 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

**Control parameters**

Exposure guidelines

| Chemical name                 | ACGIH TLV         | OSHA PEL                                                                                                                                                                                                                               | Ontario                   | European Union       |        |
|-------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------|--------|
| 1,2,3-Propanetriol<br>56-81-5 | No data available | TWA: 15 mg/m <sup>3</sup> mist, total particulate<br>TWA: 5 mg/m <sup>3</sup> mist, respirable fraction<br>(vacated) TWA: 10 mg/m <sup>3</sup> mist, total particulate<br>(vacated) TWA: 5 mg/m <sup>3</sup> mist, respirable fraction | -                         | -                    |        |
| Chemical name                 | China             | Japan Society of Occupational Health                                                                                                                                                                                                   | OEL                       | Australia            | Taiwan |
| 1,2,3-Propanetriol<br>56-81-5 | -                 | -                                                                                                                                                                                                                                      | TWA: 10 mg/m <sup>3</sup> | 10 mg/m <sup>3</sup> | -      |

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Skin and body protection** Wear suitable protective clothing.

**Hand protection** Wear suitable gloves.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

## 9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

|                |                  |                 |                          |
|----------------|------------------|-----------------|--------------------------|
| Physical state | Liquid           | Odour           | Odourless                |
| Appearance     | aqueous solution | Odour threshold | No information available |
| Colour         | white            |                 |                          |

| <u>Property</u>                              | <u>Values</u>     | <u>Remarks • Method</u>  |
|----------------------------------------------|-------------------|--------------------------|
| pH                                           |                   | No information available |
| Melting point / freezing point               |                   | No information available |
| Boiling point / boiling range                | > 100 °C / 212 °F |                          |
| Flash point                                  | > 160 °C / 320 °F |                          |
| Evaporation rate                             |                   | No information available |
| Flammability (solid, gas)                    |                   | No information available |
| Upper/lower flammability or explosive limits |                   |                          |
| Upper flammability or explosive limits       | Not applicable    |                          |
| Lower flammability or explosive limits       | Not applicable    |                          |
| Vapour pressure                              |                   | No information available |
| Vapour density                               |                   | No information available |
| Relative density                             |                   | No information available |
| Solubility(ies)                              |                   |                          |
| Water solubility                             | Miscible in water |                          |
| Solubility in other solvents                 |                   | No information available |
| Partition coefficient                        |                   | No information available |
| Autoignition temperature                     |                   | No information available |
| Decomposition temperature                    |                   | No information available |
| Viscosity                                    |                   |                          |
| Kinematic viscosity                          |                   | No information available |
| Dynamic viscosity                            |                   |                          |

Other information

|                      |                |
|----------------------|----------------|
| Oxidising properties | Not applicable |
|----------------------|----------------|

**10. STABILITY AND REACTIVITY**Reactivity

|            |                           |
|------------|---------------------------|
| Reactivity | No information available. |
|------------|---------------------------|

Chemical stability

|           |                                 |
|-----------|---------------------------------|
| Stability | Stable under normal conditions. |
|-----------|---------------------------------|

Explosion data

|                                  |       |
|----------------------------------|-------|
| Sensitivity to mechanical impact | None  |
| Sensitivity to static discharge  | None. |

Possibility of hazardous reactions

|                                    |                               |
|------------------------------------|-------------------------------|
| Possibility of hazardous reactions | None under normal processing. |
|------------------------------------|-------------------------------|

Conditions to avoid

|                     |                                           |
|---------------------|-------------------------------------------|
| Conditions to avoid | None known based on information supplied. |
|---------------------|-------------------------------------------|

Incompatible materials

|                        |                                           |
|------------------------|-------------------------------------------|
| Incompatible materials | None known based on information supplied. |
|------------------------|-------------------------------------------|

Hazardous decomposition products

**Hazardous decomposition products** None known based on information supplied.

## 11. TOXICOLOGICAL INFORMATION

### Information on the likely routes of exposure

#### Product Information

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available.

**Symptoms** No information available.

### Acute toxicity

#### Numerical measures of toxicity

0 % of the mixture consists of ingredient(s) of unknown acute oral toxicity  
 0 % of the mixture consists of ingredient(s) of unknown acute dermal toxicity  
 29.467 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (gas)  
 29.467 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (vapor)  
 0 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (dust/mist)

#### Component Information

| Chemical name      | Oral LD50             | Dermal LD50          | Inhalation LC50                     |
|--------------------|-----------------------|----------------------|-------------------------------------|
| 1,2,3-Propanetriol | = 12600 mg/kg ( Rat ) | > 10 g/kg ( Rabbit ) | > 570 mg/m <sup>3</sup> ( Rat ) 1 h |

### Delayed and immediate effects and also chronic effects from short and long term exposure

**Skin corrosion/irritation** No information available.

**Serious eye damage/irritation** No information available.

**Respiratory or skin sensitization** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Target organ effects** Kidney, Respiratory system, Eyes, Skin.

**Aspiration hazard** No information available.

## 12. ECOLOGICAL INFORMATION

### Toxicity

0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment

**Ecotoxicity** Harmful to aquatic life.

| Chemical name      | Algae/aquatic plants | Fish                                                 | Crustacea                                   |
|--------------------|----------------------|------------------------------------------------------|---------------------------------------------|
| 1,2,3-Propanetriol | -                    | LC50: 51 - 57mL/L (96h, <i>Oncorhynchus mykiss</i> ) | EC50: >500mg/L (24h, <i>Daphnia magna</i> ) |

**Persistence and degradability**

No information available.

**Bioaccumulative potential**

There is no data for this product.

**Mobility**

**Mobility in soil** No information available.

**Mobility** No information available.

| Chemical name      | Partition coefficient |
|--------------------|-----------------------|
| 1,2,3-Propanetriol | -1.76                 |

**Other adverse effects**

No information available.

### 13. DISPOSAL CONSIDERATIONS

**Disposal methods**

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

### 14. TRANSPORT INFORMATION

**IMDG** Not regulated  
**Transport in bulk according to Annex II of MARPOL and the IBC Code** No information available

**IATA** Not regulated

**RID** Not regulated

**ADR** Not regulated

**ADN** Not regulated

**Special precautions for user** Special provisions from the regulations relative to the specified mode of transport are noted by numeric code. Refer to the regulations for the full text of special provisions.

### 15. REGULATORY INFORMATION

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

The Rotterdam Convention Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**16. OTHER INFORMATION**

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 01-Sep-2021

**Revision Note** SDS sections updated. 1. 2.

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Disclaimer**

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**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
The Globally Harmonised System of Classification and Labelling of Chemicals (GHS)

Revision date 25-Aug-2021

Revision Number 1.1

## 1. IDENTIFICATION

### Product identifier

**Product Name** Conjugate

### Other means of identification

**Catalogue Number(s)** 220HSP, 220HAN, 220HDS, 220HCE

**Registration Number(s)** No information available

### Recommended use of the chemical and restrictions on use

**Recommended use** In-vitro laboratory reagent or component

### Supplier's details

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Diagnostic Group  
4000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pvt. Ltd.  
Bio-Rad House  
86-87, Udyog Vihar Phase IV Gurgaon  
122005  
Haryana India

Bio-Rad Laboratories (Pty) Ltd.  
34 Bolton Road  
Parkwood, Johannesburg 2193  
South Africa

### Technical Service

India: 91-124-4029300 or 1-800-180-1224  
South Africa: 27-11-442-85-08  
India: support.india@bio-rad.com  
South Africa: cdg\_techsupport\_eemea@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC India: 000-800-100-7141  
CHEMTREC South Africa: 0-800-983-611

## 2. HAZARDS IDENTIFICATION

### Classification of the substance or mixture

Skin sensitization

Category 1A

### GHS Label elements, including precautionary statements



**Signal word**

**Warning**

### **Hazard statements**

May cause an allergic skin reaction



**Precautionary Statements - Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves

**Precautionary Statements - Response**

Specific treatment (see .? on this label)

**Skin**

IF ON SKIN: Wash with plenty of water and soap  
If skin irritation or rash occurs: Get medical advice/attention  
Take off contaminated clothing and wash it before reuse

**Precautionary Statements - Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other hazards which do not result in classification**

No information available

**3. COMPOSITION/INFORMATION ON INGREDIENTS****Substance**

Not applicable

**Mixture**

| Chemical name | CAS No | Weight-% |
|---------------|--------|----------|
| Trade secret  | -      | 0 - 10%  |

**4. FIRST AID MEASURES****Description of necessary first aid measures**

|                       |                                                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Show this safety data sheet to the doctor in attendance.                                                                             |
| <b>Inhalation</b>     | Remove to fresh air.                                                                                                                 |
| <b>Skin contact</b>   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a physician. |
| <b>Eye contact</b>    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician.                 |
| <b>Ingestion</b>      | Rinse mouth thoroughly with water.                                                                                                   |

**For emergency responders**

|                                           |                           |
|-------------------------------------------|---------------------------|
| <b>Self-protection of the first aider</b> | No information available. |
|-------------------------------------------|---------------------------|

**Most important symptoms and effects, both acute and delayed**

|                 |                         |
|-----------------|-------------------------|
| <b>Symptoms</b> | Itching. Rashes. Hives. |
|-----------------|-------------------------|

**Indication of immediate medical attention and special treatment needed, if necessary**

|                           |                                                                        |
|---------------------------|------------------------------------------------------------------------|
| <b>Note to physicians</b> | May cause sensitization in susceptible persons. Treat symptomatically. |
|---------------------------|------------------------------------------------------------------------|

**5. FIREFIGHTING MEASURES****Suitable Extinguishing Media**

|                                     |                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|

|                   |                                                                    |
|-------------------|--------------------------------------------------------------------|
| <b>Large Fire</b> | CAUTION: Use of water spray when fighting fire may be inefficient. |
|-------------------|--------------------------------------------------------------------|

**Unsuitable extinguishing media** Do not scatter spilled material with high pressure water streams.

**Specific hazards arising from the chemical**

**Specific hazards arising from the chemical** Product is or contains a sensitizer. May cause sensitization by skin contact.

**Special protective actions for fire-fighters**

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## 6. ACCIDENTAL RELEASE MEASURES

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Environmental precautions**

**Environmental precautions** See Section 12 for additional Ecological Information.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## 7. HANDLING AND STORAGE

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children.

## 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

**Control parameters**

**Exposure guidelines** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Wear safety glasses with side shields (or goggles).

|                                       |                                                                                                                                                                          |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>       | Wear suitable protective clothing.                                                                                                                                       |
| <b>Hand protection</b>                | Wear suitable gloves.                                                                                                                                                    |
| <b>Respiratory protection</b>         | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| <b>General hygiene considerations</b> | Handle in accordance with good industrial hygiene and safety practice.                                                                                                   |

## 9. PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                                     |                   |                          |                          |
|-----------------------------------------------------|-------------------|--------------------------|--------------------------|
| <b>Physical state</b>                               | Liquid            | <b>Odour</b>             | Odourless                |
| <b>Appearance</b>                                   | aqueous solution  | <b>Odour threshold</b>   | No information available |
| <b>Colour</b>                                       | amber             |                          |                          |
| <b>Property</b>                                     | <b>Values</b>     | <b>Remarks • Method</b>  |                          |
| <b>pH</b>                                           | 7.2               |                          |                          |
| <b>Melting point / freezing point</b>               |                   | No information available |                          |
| <b>Boiling point / boiling range</b>                | 100 °C / 212 °F   |                          |                          |
| <b>Flash point</b>                                  |                   | No information available |                          |
| <b>Evaporation rate</b>                             |                   | No information available |                          |
| <b>Flammability (solid, gas)</b>                    |                   | No information available |                          |
| <b>Upper/lower flammability or explosive limits</b> |                   |                          |                          |
| <b>Upper flammability or explosive limits</b>       | Not applicable    |                          |                          |
| <b>Lower flammability or explosive limits</b>       | Not applicable    |                          |                          |
| <b>Vapour pressure</b>                              |                   | No information available |                          |
| <b>Vapour density</b>                               |                   | No information available |                          |
| <b>Relative density</b>                             |                   | No information available |                          |
| <b>Solubility(ies)</b>                              |                   |                          |                          |
| <b>Water solubility</b>                             | Miscible in water |                          |                          |
| <b>Solubility in other solvents</b>                 |                   | No information available |                          |
| <b>Partition coefficient</b>                        |                   | No information available |                          |
| <b>Autoignition temperature</b>                     |                   | No information available |                          |
| <b>Decomposition temperature</b>                    |                   | No information available |                          |
| <b>Viscosity</b>                                    |                   |                          |                          |
| <b>Kinematic viscosity</b>                          |                   | No information available |                          |
| <b>Dynamic viscosity</b>                            |                   |                          |                          |

### Other information

|                             |                |
|-----------------------------|----------------|
| <b>Oxidising properties</b> | Not applicable |
|-----------------------------|----------------|

## 10. STABILITY AND REACTIVITY

### Reactivity

|                   |                           |
|-------------------|---------------------------|
| <b>Reactivity</b> | No information available. |
|-------------------|---------------------------|

### Chemical stability

|                  |                                 |
|------------------|---------------------------------|
| <b>Stability</b> | Stable under normal conditions. |
|------------------|---------------------------------|

### Explosion data

|                                         |       |
|-----------------------------------------|-------|
| <b>Sensitivity to mechanical impact</b> | None  |
| <b>Sensitivity to static discharge</b>  | None. |

### Possibility of hazardous reactions

|                                           |                               |
|-------------------------------------------|-------------------------------|
| <b>Possibility of hazardous reactions</b> | None under normal processing. |
|-------------------------------------------|-------------------------------|

**Conditions to avoid**

**Conditions to avoid** None known based on information supplied.

**Incompatible materials**

**Incompatible materials** None known based on information supplied.

**Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

**11. TOXICOLOGICAL INFORMATION****Information on the likely routes of exposure****Product Information**

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** May cause sensitization by skin contact. Specific test data for the substance or mixture is not available. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons. (based on components).

**Ingestion** Specific test data for the substance or mixture is not available.

**Symptoms** Itching. Rashes. Hives.

**Acute toxicity****Numerical measures of toxicity**

0.94151 % of the mixture consists of ingredient(s) of unknown acute oral toxicity

0.94151 % of the mixture consists of ingredient(s) of unknown acute dermal toxicity

0.94151 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (gas)

0.94151 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (vapor)

0.94151 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (dust/mist)

**Component Information**

| Chemical name | Oral LD50                                      | Dermal LD50            | Inhalation LC50         |
|---------------|------------------------------------------------|------------------------|-------------------------|
| Trade secret  | 232 - 249 mg/kg ( Rat )<br>= 120 mg/kg ( Rat ) | = 200 mg/kg ( Rabbit ) | = 0.11 mg/L ( Rat ) 4 h |

**Delayed and immediate effects and also chronic effects from short and long term exposure**

**Skin corrosion/irritation** No information available.

**Serious eye damage/irritation** No information available.

**Respiratory or skin sensitization** May cause sensitization by skin contact.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

STOT - repeated exposure No information available.

Aspiration hazard No information available.

## 12. ECOLOGICAL INFORMATION

### Toxicity

0.94151 % of the mixture consists of component(s) of unknown hazards to the aquatic environment

### Ecotoxicity

### Persistence and degradability

No information available.

### Bioaccumulative potential

There is no data for this product.

### Mobility

Mobility in soil No information available.

Mobility No information available.

### Other adverse effects

No information available.

## 13. DISPOSAL CONSIDERATIONS

### Disposal methods

Waste from residues/unused products Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

Contaminated packaging Do not reuse empty containers.

## 14. TRANSPORT INFORMATION

IMDG Not regulated  
Transport in bulk according to Annex II of MARPOL and the IBC Code No information available

IATA Not regulated

RID Not regulated

ADR Not regulated

ADN Not regulated

Special precautions for user Special provisions from the regulations relative to the specified mode of transport are noted by numeric code. Refer to the regulations for the full text of special provisions.

## 15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**16. OTHER INFORMATION**

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 25-Aug-2021

**Revision Note** SDS sections updated. 1.

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
The Globally Harmonised System of Classification and Labelling of Chemicals (GHS)

Revision date 27-Aug-2021

Revision Number 1.1

## 1. IDENTIFICATION

### Product identifier

**Product Name** Wash Concentrate

### Other means of identification

**Catalogue Number(s)** 230AW

**Registration Number(s)** No information available

### Recommended use of the chemical and restrictions on use

**Recommended use** In-vitro laboratory reagent or component

### Supplier's details

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Diagnostic Group  
4000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pvt. Ltd.  
Bio-Rad House  
86-87, Udyog Vihar Phase IV Gurgaon  
122005  
Haryana India

Bio-Rad Laboratories (Pty) Ltd.  
34 Bolton Road  
Parkwood, Johannesburg 2193  
South Africa

### Technical Service

India: 91-124-4029300 or 1-800-180-1224  
South Africa: 27-11-442-85-08  
India: support.india@bio-rad.com  
South Africa: cdg\_techsupport\_eemea@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC India: 000-800-100-7141  
CHEMTREC South Africa: 0-800-983-611

## 2. HAZARDS IDENTIFICATION

### Classification of the substance or mixture

### GHS Label elements, including precautionary statements

### Other hazards which do not result in classification

No information available

## 3. COMPOSITION/INFORMATION ON INGREDIENTS

### Substance

Not applicable

### Mixture

| Chemical name                                     | CAS No    | Weight-% |
|---------------------------------------------------|-----------|----------|
| Water<br>7732-18-5                                | 7732-18-5 | 84.962   |
| Sodium chloride<br>7647-14-5                      | 7647-14-5 | 11.682   |
| Sodium phosphate dibasic<br>7558-79-4             | 7558-79-4 | 1.606    |
| Polyoxyethylene sorbitan monolaurate<br>9005-64-5 | 9005-64-5 | 1.41     |
| Phosphoric acid, monosodium salt<br>7558-80-7     | 7558-80-7 | 0.34     |

#### 4. FIRST AID MEASURES

##### Description of necessary first aid measures

|              |                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                 |
| Skin contact | Wash skin with soap and water. In the case of skin irritation or allergic reactions see a physician.                 |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| Ingestion    | Rinse mouth thoroughly with water.                                                                                   |

##### For emergency responders

|                                    |                           |
|------------------------------------|---------------------------|
| Self-protection of the first aider | No information available. |
|------------------------------------|---------------------------|

##### Most important symptoms and effects, both acute and delayed

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

##### Indication of immediate medical attention and special treatment needed, if necessary

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

#### 5. FIREFIGHTING MEASURES

##### Suitable Extinguishing Media

|                              |                                                                                                         |
|------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|------------------------------|---------------------------------------------------------------------------------------------------------|

|            |                                                                    |
|------------|--------------------------------------------------------------------|
| Large Fire | CAUTION: Use of water spray when fighting fire may be inefficient. |
|------------|--------------------------------------------------------------------|

|                                |                                                                   |
|--------------------------------|-------------------------------------------------------------------|
| Unsuitable extinguishing media | Do not scatter spilled material with high pressure water streams. |
|--------------------------------|-------------------------------------------------------------------|

##### Specific hazards arising from the chemical

|                                            |                           |
|--------------------------------------------|---------------------------|
| Specific hazards arising from the chemical | No information available. |
|--------------------------------------------|---------------------------|

##### Special protective actions for fire-fighters

|                                                                |                                                                                                                                    |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|

#### 6. ACCIDENTAL RELEASE MEASURES

##### Personal precautions, protective equipment and emergency procedures



**Personal precautions** Ensure adequate ventilation.

#### Environmental precautions

**Environmental precautions** See Section 12 for additional Ecological Information.

#### Methods and material for containment and cleaning up

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## 7. HANDLING AND STORAGE

#### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

#### Conditions for safe storage, including any incompatibilities

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place.

## 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

#### Control parameters

**Exposure guidelines** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies

#### Appropriate engineering controls

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

#### Individual protection measures, such as personal protective equipment

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Skin and body protection** Wear suitable protective clothing.

**Hand protection** Wear suitable gloves.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

## 9. PHYSICAL AND CHEMICAL PROPERTIES

#### Information on basic physical and chemical properties

|                                       |                   |                          |                          |
|---------------------------------------|-------------------|--------------------------|--------------------------|
| <b>Physical state</b>                 | Liquid            | <b>Odour</b>             | Odourless                |
| <b>Appearance</b>                     | aqueous solution  | <b>Odour threshold</b>   | No information available |
| <b>Colour</b>                         | white             |                          |                          |
| <b>Property</b>                       | <b>Values</b>     | <b>Remarks • Method</b>  |                          |
| <b>pH</b>                             | 7.3               |                          |                          |
| <b>Melting point / freezing point</b> |                   | No information available |                          |
| <b>Boiling point / boiling range</b>  | > 100 °C / 212 °F |                          |                          |
| <b>Flash point</b>                    |                   | No information available |                          |

|                                              |                          |
|----------------------------------------------|--------------------------|
| Evaporation rate                             | No information available |
| Flammability (solid, gas)                    | No information available |
| Upper/lower flammability or explosive limits |                          |
| Upper flammability or explosive limits       | Not applicable           |
| Lower flammability or explosive limits       | Not applicable           |
| Vapour pressure                              | No information available |
| Vapour density                               | No information available |
| Relative density                             | No information available |
| Solubility(ies)                              |                          |
| Water solubility                             | Miscible in water        |
| Solubility in other solvents                 | No information available |
| Partition coefficient                        | No information available |
| Autoignition temperature                     | No information available |
| Decomposition temperature                    | No information available |
| Viscosity                                    |                          |
| Kinematic viscosity                          | No information available |
| Dynamic viscosity                            |                          |

**Other information**

|                      |                |
|----------------------|----------------|
| Oxidising properties | Not applicable |
|----------------------|----------------|

**10. STABILITY AND REACTIVITY****Reactivity**

|            |                           |
|------------|---------------------------|
| Reactivity | No information available. |
|------------|---------------------------|

**Chemical stability**

|           |                                 |
|-----------|---------------------------------|
| Stability | Stable under normal conditions. |
|-----------|---------------------------------|

**Explosion data**

|                                  |       |
|----------------------------------|-------|
| Sensitivity to mechanical impact | None  |
| Sensitivity to static discharge  | None. |

**Possibility of hazardous reactions**

|                                    |                               |
|------------------------------------|-------------------------------|
| Possibility of hazardous reactions | None under normal processing. |
|------------------------------------|-------------------------------|

**Conditions to avoid**

|                     |                                           |
|---------------------|-------------------------------------------|
| Conditions to avoid | None known based on information supplied. |
|---------------------|-------------------------------------------|

**Incompatible materials**

|                        |                                           |
|------------------------|-------------------------------------------|
| Incompatible materials | None known based on information supplied. |
|------------------------|-------------------------------------------|

**Hazardous decomposition products**

|                                  |                                           |
|----------------------------------|-------------------------------------------|
| Hazardous decomposition products | None known based on information supplied. |
|----------------------------------|-------------------------------------------|

**11. TOXICOLOGICAL INFORMATION****Information on the likely routes of exposure****Product Information**

|             |                                                                   |
|-------------|-------------------------------------------------------------------|
| Inhalation  | Specific test data for the substance or mixture is not available. |
| Eye contact | Specific test data for the substance or mixture is not available. |

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available.

**Symptoms** No information available.

#### Acute toxicity

#### Numerical measures of toxicity

0 % of the mixture consists of ingredient(s) of unknown acute oral toxicity  
 1.41 % of the mixture consists of ingredient(s) of unknown acute dermal toxicity  
 14.698 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (gas)  
 14.698 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (vapor)  
 1.41 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (dust/mist)

The following values are calculated based on chapter 3.1 of the GHS document

**ATEmix (oral)** 25,428.90 mg/kg

| Chemical name                        | Oral LD50                                      | Dermal LD50             | Inhalation LC50                   |
|--------------------------------------|------------------------------------------------|-------------------------|-----------------------------------|
| Water                                | > 90 mL/kg ( Rat )                             | -                       | -                                 |
| Sodium chloride                      | = 3 g/kg ( Rat )                               | > 10 g/kg ( Rabbit )    | > 42 g/m <sup>3</sup> ( Rat ) 1 h |
| Sodium phosphate dibasic             | = 17 g/kg ( Rat )                              | -                       | -                                 |
| Polyoxyethylene sorbitan monolaurate | = 37000 mg/kg ( Rat )<br>= 36700 µL/kg ( Rat ) | -                       | -                                 |
| Phosphoric acid, monosodium salt     | = 8290 mg/kg ( Rat )                           | > 7940 mg/kg ( Rabbit ) | -                                 |

#### Delayed and immediate effects and also chronic effects from short and long term exposure

**Skin corrosion/irritation** No information available.

**Serious eye damage/irritation** No information available.

**Respiratory or skin sensitization** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Aspiration hazard** No information available.

## 12. ECOLOGICAL INFORMATION

#### Toxicity

0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment

#### Ecotoxicity

| Chemical name   | Algae/aquatic plants | Fish                        | Crustacea                     |
|-----------------|----------------------|-----------------------------|-------------------------------|
| Sodium chloride | -                    | LC50: 4747 - 7824mg/L (96h, | EC50: 340.7 - 469.2mg/L (48h, |

|  |  |                                                                                                                                                                                                                                                                                                                                                          |                                                                                |
|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
|  |  | <i>Oncorhynchus mykiss</i><br>LC50: 5560 - 6080mg/L (96h,<br><i>Lepomis macrochirus</i> )<br>LC50: 6020 - 7070mg/L (96h,<br><i>Pimephales promelas</i> )<br>LC50: 6420 - 6700mg/L (96h,<br><i>Pimephales promelas</i> )<br>LC50: =12946mg/L (96h, <i>Lepomis</i><br><i>macrochirus</i> )<br>LC50: =7050mg/L (96h, <i>Pimephales</i><br><i>promelas</i> ) | <i>Daphnia magna</i><br>EC50: =1000mg/L (48h, <i>Daphnia</i><br><i>magna</i> ) |
|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|

**Persistence and degradability**

No information available.

**Bioaccumulative potential**

No information available.

**Mobility****Mobility in soil** No information available.**Mobility** No information available.**Other adverse effects**

No information available.

**13. DISPOSAL CONSIDERATIONS****Disposal methods****Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.**Contaminated packaging** Do not reuse empty containers.**14. TRANSPORT INFORMATION**
**IMDG** Not regulated  
**Transport in bulk according to** No information available  
**Annex II of MARPOL and the IBC**  
**Code**
**IATA** Not regulated**RID** Not regulated**ADR** Not regulated**ADN** Not regulated**Special precautions for user** Special provisions from the regulations relative to the specified mode of transport are noted by numeric code. Refer to the regulations for the full text of special provisions.**15. REGULATORY INFORMATION****Safety, health and environmental regulations/legislation specific for the substance or mixture****International Regulations****The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

**International Inventories**

Contact supplier for inventory compliance status

## 16. OTHER INFORMATION

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 27-Aug-2021

**Revision Note** SDS sections updated. 1. 2.

**Key or legend to abbreviations and acronyms used in the safety data sheet**

**Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
The Globally Harmonised System of Classification and Labelling of Chemicals (GHS)

Revision date 01-Sep-2021

Revision Number 1

## 1. IDENTIFICATION

### Product identifier

**Product Name** Sample Diluent

### Other means of identification

**Catalogue Number(s)** 230AD

**Registration Number(s)** No information available

### Recommended use of the chemical and restrictions on use

**Recommended use** In-vitro laboratory reagent or component

### Supplier's details

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Diagnostic Group  
4000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pvt. Ltd.  
Bio-Rad House  
86-87, Udyog Vihar Phase IV Gurgaon  
122005  
Haryana India

Bio-Rad Laboratories (Pty) Ltd.  
34 Bolton Road  
Parkwood, Johannesburg 2193  
South Africa

### Technical Service

India: 91-124-4029300 or 1-800-180-1224  
South Africa: 27-11-442-85-08  
India: support.india@bio-rad.com  
South Africa: cdg\_techsupport\_eemea@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC India: 000-800-100-7141  
CHEMTREC South Africa: 0-800-983-611

## 2. HAZARDS IDENTIFICATION

### Classification of the substance or mixture

### GHS Label elements, including precautionary statements

### Other hazards which do not result in classification

No information available

## 3. COMPOSITION/INFORMATION ON INGREDIENTS

### Substance

Not applicable

### Mixture

| Chemical name                | CAS No    | Weight-% |
|------------------------------|-----------|----------|
| Sodium chloride<br>7647-14-5 | 7647-14-5 | 1.206    |

#### 4. FIRST AID MEASURES

##### Description of necessary first aid measures

|              |                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                 |
| Skin contact | Wash skin with soap and water. In the case of skin irritation or allergic reactions see a physician.                 |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| Ingestion    | Rinse mouth thoroughly with water.                                                                                   |

##### For emergency responders

|                                    |                           |
|------------------------------------|---------------------------|
| Self-protection of the first aider | No information available. |
|------------------------------------|---------------------------|

##### Most important symptoms and effects, both acute and delayed

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

##### Indication of immediate medical attention and special treatment needed, if necessary

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

#### 5. FIREFIGHTING MEASURES

##### Suitable Extinguishing Media

|                              |                                                                                                         |
|------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|------------------------------|---------------------------------------------------------------------------------------------------------|

|            |                                                                    |
|------------|--------------------------------------------------------------------|
| Large Fire | CAUTION: Use of water spray when fighting fire may be inefficient. |
|------------|--------------------------------------------------------------------|

|                                |                                                                   |
|--------------------------------|-------------------------------------------------------------------|
| Unsuitable extinguishing media | Do not scatter spilled material with high pressure water streams. |
|--------------------------------|-------------------------------------------------------------------|

##### Specific hazards arising from the chemical

|                                            |                           |
|--------------------------------------------|---------------------------|
| Specific hazards arising from the chemical | No information available. |
|--------------------------------------------|---------------------------|

##### Special protective actions for fire-fighters

|                                                                |                                                                                                                                    |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|

#### 6. ACCIDENTAL RELEASE MEASURES

##### Personal precautions, protective equipment and emergency procedures

|                      |                              |
|----------------------|------------------------------|
| Personal precautions | Ensure adequate ventilation. |
|----------------------|------------------------------|

##### Environmental precautions

|                           |                                                       |
|---------------------------|-------------------------------------------------------|
| Environmental precautions | See Section 12 for additional Ecological Information. |
|---------------------------|-------------------------------------------------------|

##### Methods and material for containment and cleaning up

|                                        |                                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------|
| <b>Methods for containment</b>         | Prevent further leakage or spillage if safe to do so.                                |
| <b>Methods for cleaning up</b>         | Pick up and transfer to properly labeled containers.                                 |
| <b>Prevention of secondary hazards</b> | Clean contaminated objects and areas thoroughly observing environmental regulations. |

## 7. HANDLING AND STORAGE

### Precautions for safe handling

|                                |                                                                        |
|--------------------------------|------------------------------------------------------------------------|
| <b>Advice on safe handling</b> | Handle in accordance with good industrial hygiene and safety practice. |
|--------------------------------|------------------------------------------------------------------------|

### Conditions for safe storage, including any incompatibilities

|                               |                                                                          |
|-------------------------------|--------------------------------------------------------------------------|
| <b>Storage Conditions</b>     | Keep containers tightly closed in a dry, cool and well-ventilated place. |
| <b>Incompatible materials</b> | Metals.                                                                  |

## 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control parameters

|                            |                                                                                                                                                            |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure guidelines</b> | This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Appropriate engineering controls

|                             |                                                     |
|-----------------------------|-----------------------------------------------------|
| <b>Engineering controls</b> | Showers<br>Eyewash stations<br>Ventilation systems. |
|-----------------------------|-----------------------------------------------------|

### Individual protection measures, such as personal protective equipment

|                                       |                                                                                                                                                                          |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face protection</b>            | Wear safety glasses with side shields (or goggles).                                                                                                                      |
| <b>Skin and body protection</b>       | Wear suitable protective clothing.                                                                                                                                       |
| <b>Hand protection</b>                | Wear suitable gloves.                                                                                                                                                    |
| <b>Respiratory protection</b>         | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| <b>General hygiene considerations</b> | Handle in accordance with good industrial hygiene and safety practice.                                                                                                   |

## 9. PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                              |                   |                          |                          |
|----------------------------------------------|-------------------|--------------------------|--------------------------|
| <b>Physical state</b>                        | Liquid            | <b>Odour</b>             | Odourless                |
| <b>Appearance</b>                            | aqueous solution  | <b>Odour threshold</b>   | No information available |
| <b>Colour</b>                                | white             |                          |                          |
| <b>Property</b>                              | <b>Values</b>     | <b>Remarks • Method</b>  |                          |
| pH                                           | 7.3               | No information available |                          |
| Melting point / freezing point               |                   | No information available |                          |
| Boiling point / boiling range                | > 100 °C / 212 °F | No information available |                          |
| Flash point                                  |                   | No information available |                          |
| Evaporation rate                             |                   | No information available |                          |
| Flammability (solid, gas)                    |                   | No information available |                          |
| Upper/lower flammability or explosive limits |                   |                          |                          |
| Upper flammability or explosive limits       | Not applicable    |                          |                          |
| Lower flammability or explosive              | Not applicable    |                          |                          |



|                              |                          |
|------------------------------|--------------------------|
| limits                       |                          |
| Vapour pressure              | No information available |
| Vapour density               | No information available |
| Relative density             | No information available |
| Solubility(ies)              |                          |
| Water solubility             | Miscible in water        |
| Solubility in other solvents | No information available |
| Partition coefficient        | No information available |
| Autoignition temperature     | No information available |
| Decomposition temperature    | No information available |
| Viscosity                    |                          |
| Kinematic viscosity          | No information available |
| Dynamic viscosity            |                          |

**Other information**

|                      |                |
|----------------------|----------------|
| Oxidising properties | Not applicable |
|----------------------|----------------|

**10. STABILITY AND REACTIVITY****Reactivity**

|            |                           |
|------------|---------------------------|
| Reactivity | No information available. |
|------------|---------------------------|

**Chemical stability**

|           |                                 |
|-----------|---------------------------------|
| Stability | Stable under normal conditions. |
|-----------|---------------------------------|

**Explosion data**

|                                  |       |
|----------------------------------|-------|
| Sensitivity to mechanical impact | None  |
| Sensitivity to static discharge  | None. |

**Possibility of hazardous reactions**

|                                    |                               |
|------------------------------------|-------------------------------|
| Possibility of hazardous reactions | None under normal processing. |
|------------------------------------|-------------------------------|

**Conditions to avoid**

|                     |                                           |
|---------------------|-------------------------------------------|
| Conditions to avoid | None known based on information supplied. |
|---------------------|-------------------------------------------|

**Incompatible materials**

|                        |         |
|------------------------|---------|
| Incompatible materials | Metals. |
|------------------------|---------|

**Hazardous decomposition products**

|                                  |                                           |
|----------------------------------|-------------------------------------------|
| Hazardous decomposition products | None known based on information supplied. |
|----------------------------------|-------------------------------------------|

**11. TOXICOLOGICAL INFORMATION****Information on the likely routes of exposure****Product Information**

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |
| Symptoms     | No information available.                                         |

Acute toxicity**Numerical measures of toxicity**

0 % of the mixture consists of ingredient(s) of unknown acute oral toxicity  
 0 % of the mixture consists of ingredient(s) of unknown acute dermal toxicity  
 1.206 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (gas)  
 1.206 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (vapor)  
 0 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (dust/mist)

| Chemical name   | Oral LD50        | Dermal LD50          | Inhalation LC50                   |
|-----------------|------------------|----------------------|-----------------------------------|
| Sodium chloride | = 3 g/kg ( Rat ) | > 10 g/kg ( Rabbit ) | > 42 g/m <sup>3</sup> ( Rat ) 1 h |

**Delayed and immediate effects and also chronic effects from short and long term exposure**

|                                          |                           |
|------------------------------------------|---------------------------|
| <b>Skin corrosion/irritation</b>         | No information available. |
| <b>Serious eye damage/irritation</b>     | No information available. |
| <b>Respiratory or skin sensitization</b> | No information available. |
| <b>Germ cell mutagenicity</b>            | No information available. |
| <b>Carcinogenicity</b>                   | No information available. |
| <b>Reproductive toxicity</b>             | No information available. |
| <b>STOT - single exposure</b>            | No information available. |
| <b>STOT - repeated exposure</b>          | No information available. |
| <b>Aspiration hazard</b>                 | No information available. |

**12. ECOLOGICAL INFORMATION**Toxicity

0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment

**Ecotoxicity**

| Chemical name   | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                                                                                      | Crustacea                                                                                            |
|-----------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Sodium chloride | -                    | LC50: 4747 - 7824mg/L (96h, <i>Oncorhynchus mykiss</i> )<br>LC50: 5560 - 6080mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: 6020 - 7070mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 6420 - 6700mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: =12946mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =7050mg/L (96h, <i>Pimephales promelas</i> ) | EC50: 340.7 - 469.2mg/L (48h, <i>Daphnia magna</i> )<br>EC50: =1000mg/L (48h, <i>Daphnia magna</i> ) |

**Persistence and degradability**

No information available.

**Bioaccumulative potential**

No information available.

**Mobility**

**Mobility in soil** No information available.

**Mobility** No information available.

**Other adverse effects**

No information available.

**13. DISPOSAL CONSIDERATIONS****Disposal methods**

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

**14. TRANSPORT INFORMATION**

**IMDG** Not regulated  
**Transport in bulk according to Annex II of MARPOL and the IBC Code** No information available

**IATA** Not regulated

**RID** Not regulated

**ADR** Not regulated

**ADN** Not regulated

**Special precautions for user** Special provisions from the regulations relative to the specified mode of transport are noted by numeric code. Refer to the regulations for the full text of special provisions.

**15. REGULATORY INFORMATION****Safety, health and environmental regulations/legislation specific for the substance or mixture****International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**16. OTHER INFORMATION**

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 01-Sep-2021

**Revision Note** SDS sections updated. 1. 2.

**Key or legend to abbreviations and acronyms used in the safety data sheet**

**Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

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**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
The Globally Harmonised System of Classification and Labelling of Chemicals (GHS)

Revision date 25-Aug-2021

Revision Number 1.1

## 1. IDENTIFICATION

### Product identifier

Product Name Substrate

### Other means of identification

Catalogue Number(s) 220TM

UN/ID no UN1987

Registration Number(s) No information available

### Recommended use of the chemical and restrictions on use

Recommended use In-vitro laboratory reagent or component

### Supplier's details

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Diagnostic Group  
4000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pvt. Ltd.  
Bio-Rad House  
86-87, Udyog Vihar Phase IV Gurgaon  
122005  
Haryana India

Bio-Rad Laboratories (Pty) Ltd.  
34 Bolton Road  
Parkwood, Johannesburg 2193  
South Africa

### Technical Service

India: 91-124-4029300 or 1-800-180-1224  
South Africa: 27-11-442-85-08  
India: support.india@bio-rad.com  
South Africa: cdg\_techsupport\_eemea@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC India: 000-800-100-7141  
CHEMTREC South Africa: 0-800-983-611

## 2. HAZARDS IDENTIFICATION

### Classification of the substance or mixture

|                                                  |            |
|--------------------------------------------------|------------|
| Acute toxicity - Oral                            | Category 4 |
| Acute toxicity - Dermal                          | Category 4 |
| Acute toxicity - Inhalation (Dusts/Mists)        | Category 4 |
| Specific target organ toxicity (single exposure) | Category 1 |

### GHS Label elements, including precautionary statements



## Signal word

Danger

## Hazard statements

Harmful if swallowed  
Harmful in contact with skin  
Harmful if inhaled  
Causes damage to organs

## Precautionary Statements - Prevention

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Wear protective gloves/protective clothing/eye protection/face protection  
Use only outdoors or in a well-ventilated area  
Do not breathe dust/fume/gas/mist/vapors/spray

## Precautionary Statements - Response

Specific treatment (see .? on this label)  
IF exposed or concerned: Call a POISON CENTER or doctor

## Skin

IF ON SKIN: Wash with plenty of water and soap  
Call a POISON CENTER or doctor if you feel unwell  
Take off contaminated clothing and wash it before reuse

## Inhalation

IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Call a POISON CENTER or doctor if you feel unwell

## Ingestion

IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell  
Rinse mouth

## Precautionary Statements - Storage

Store locked up

## Precautionary Statements - Disposal

Dispose of contents/container to an approved waste disposal plant

Other hazards which do not result in classification

No information available

**3. COMPOSITION/INFORMATION ON INGREDIENTS**Substance

Not applicable

Mixture

| Chemical name       | CAS No  | Weight-% |
|---------------------|---------|----------|
| Methanol<br>67-56-1 | 67-56-1 | 15       |
| Acetone<br>67-64-1  | 67-64-1 | 10       |

**4. FIRST AID MEASURES**Description of necessary first aid measures

## General advice

Show this safety data sheet to the doctor in attendance.

## Inhalation

Remove to fresh air. IF exposed or concerned: Get medical advice/attention. If symptoms persist, call a physician. If breathing has stopped, give artificial respiration. Get medical attention immediately.

## Skin contact

Wash off immediately with plenty of water for at least 15 minutes. If symptoms persist, call a physician.

## Eye contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. If

symptoms persist, call a physician.

**Ingestion**

Do NOT induce vomiting. Rinse mouth. Never give anything by mouth to an unconscious person. Get medical attention.

**For emergency responders****Self-protection of the first aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. Avoid breathing vapors or mists. Use personal protective equipment as required. See section 8 for more information.

**Most important symptoms and effects, both acute and delayed****Symptoms**

Coughing and/ or wheezing. Difficulty in breathing.

**Indication of immediate medical attention and special treatment needed, if necessary****Note to physicians**

Treat symptomatically.

## 5. FIREFIGHTING MEASURES

**Suitable Extinguishing Media****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Large Fire**

CAUTION: Use of water spray when fighting fire may be inefficient.

**Unsuitable extinguishing media**

Do not scatter spilled material with high pressure water streams.

**Specific hazards arising from the chemical****Specific hazards arising from the chemical**

No information available.

**Special protective actions for fire-fighters****Special protective equipment and precautions for fire-fighters**

Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## 6. ACCIDENTAL RELEASE MEASURES

**Personal precautions, protective equipment and emergency procedures****Personal precautions**

Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid breathing vapors or mists.

**Environmental precautions****Environmental precautions**

Prevent further leakage or spillage if safe to do so.

**Methods and material for containment and cleaning up****Methods for containment**

Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up**

Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations.

**Other information**

Refer to protective measures listed in Sections 7 and 8.

## 7. HANDLING AND STORAGE

**Precautions for safe handling****Advice on safe handling**

Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Avoid breathing vapors or mists. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product.

**Conditions for safe storage, including any incompatibilities****Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep out of the reach of children. Store locked up.

**8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control parameters****Exposure guidelines**

| Chemical name       | ACGIH TLV                                | OSHA PEL                                                                                                                                                                                                                                                                      |                                        | Ontario                                                   | European Union                                                                |
|---------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------|
| Methanol<br>67-56-1 | STEL: 250 ppm<br>TWA: 200 ppm<br>S*      | TWA: 200 ppm<br>TWA: 260 mg/m³<br>(vacated) TWA: 200 ppm<br>(vacated) TWA: 260 mg/m³<br>(vacated) STEL: 250 ppm<br>(vacated) STEL: 325 mg/m³<br>(vacated) S*                                                                                                                  |                                        | TWA: 200 ppm<br>STEL: 250 ppm<br>Skin                     | TWA: 200 ppm<br>TWA: 260 mg/m³<br>*                                           |
| Acetone<br>67-64-1  | STEL: 500 ppm<br>TWA: 250 ppm            | TWA: 1000 ppm<br>TWA: 2400 mg/m³<br>(vacated) TWA: 750 ppm<br>(vacated) TWA: 1800 mg/m³<br>(vacated) STEL: 2400 mg/m³<br>The acetone STEL does not<br>apply to the cellulose<br>acetate fiber industry. It is in<br>effect for all other sectors.<br>(vacated) STEL: 1000 ppm |                                        | TWA: 250 ppm<br>STEL: 500 ppm                             | TWA: 500 ppm<br>TWA: 1210 mg/m³                                               |
| Chemical name       | China                                    | Japan Society of<br>Occupational Health                                                                                                                                                                                                                                       | OEL                                    | Australia                                                 | Taiwan                                                                        |
| Methanol<br>67-56-1 | TWA: 25 mg/m³<br>STEL: 50 mg/m³<br>Skin* | TWA: 200 ppm<br>TWA: 260 mg/m³<br>S*                                                                                                                                                                                                                                          | TWA: 200 ppm<br>STEL: 250 ppm<br>Skin* | 200 ppm<br>262 mg/m³<br>250 ppm STEL<br>328 mg/m³ STEL    | TWA: 200 ppm<br>TWA: 262 mg/m³<br>STEL: 250 ppm<br>STEL: 327.5 mg/m³<br>Skin* |
| Acetone<br>67-64-1  | TWA: 300 mg/m³<br>STEL: 450 mg/m³        | TWA: 200 ppm<br>TWA: 470 mg/m³                                                                                                                                                                                                                                                | TWA: 500 ppm<br>STEL: 750 ppm          | 500 ppm<br>1185 mg/m³<br>1000 ppm STEL<br>2375 mg/m³ STEL | TWA: 200 ppm<br>TWA: 475 mg/m³<br>STEL: 250 ppm<br>STEL: 593.75 mg/m³         |

**Appropriate engineering controls****Engineering controls**

Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment****Eye/face protection**

Wear safety glasses with side shields (or goggles).

**Skin and body protection**

Wear suitable protective clothing. Long sleeved clothing.

**Hand protection**

Wear suitable gloves.

**Respiratory protection**

No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations**

Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product.



## 9. PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                |        |                 |                          |
|----------------|--------|-----------------|--------------------------|
| Physical state | Liquid | Odour           | Alcohol                  |
| Appearance     | Liquid | Odour threshold | No information available |
| Colour         | white  |                 |                          |

| <u>Property</u>                              | <u>Values</u>     | <u>Remarks • Method</u>  |
|----------------------------------------------|-------------------|--------------------------|
| pH                                           |                   | No information available |
| Melting point / freezing point               |                   | No information available |
| Boiling point / boiling range                | 55.8-56.6 °C / °F |                          |
| Flash point                                  | 16 °C / 60.8 °F   |                          |
| Evaporation rate                             |                   | No information available |
| Flammability (solid, gas)                    |                   | No information available |
| Upper/lower flammability or explosive limits |                   |                          |
| Upper flammability or explosive limits       | Not applicable    |                          |
| Lower flammability or explosive limits       | Not applicable    |                          |
| Vapour pressure                              |                   | No information available |
| Vapour density                               |                   | No information available |
| Relative density                             |                   | No information available |
| Solubility(ies)                              |                   |                          |
| Water solubility                             | Miscible in water |                          |
| Solubility in other solvents                 |                   | No information available |
| Partition coefficient                        |                   | No information available |
| Autoignition temperature                     |                   | No information available |
| Decomposition temperature                    |                   | No information available |
| Viscosity                                    |                   |                          |
| Kinematic viscosity                          |                   | No information available |
| Dynamic viscosity                            |                   |                          |

### Other information

|                      |                |
|----------------------|----------------|
| Oxidising properties | Not applicable |
| Liquid Density       | 0.93909        |

## 10. STABILITY AND REACTIVITY

### Reactivity

|            |                           |
|------------|---------------------------|
| Reactivity | No information available. |
|------------|---------------------------|

### Chemical stability

|           |                                 |
|-----------|---------------------------------|
| Stability | Stable under normal conditions. |
|-----------|---------------------------------|

### Explosion data

|                                  |       |
|----------------------------------|-------|
| Sensitivity to mechanical impact | None  |
| Sensitivity to static discharge  | None. |

### Possibility of hazardous reactions

|                                    |                               |
|------------------------------------|-------------------------------|
| Possibility of hazardous reactions | None under normal processing. |
|------------------------------------|-------------------------------|

### Conditions to avoid

|                     |                 |
|---------------------|-----------------|
| Conditions to avoid | Excessive heat. |
|---------------------|-----------------|

### Incompatible materials

|                        |                                           |
|------------------------|-------------------------------------------|
| Incompatible materials | None known based on information supplied. |
|------------------------|-------------------------------------------|

**Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

## 11. TOXICOLOGICAL INFORMATION

**Information on the likely routes of exposure****Product Information**

|                     |                                                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available. Harmful by inhalation. (based on components). |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available.                                               |
| <b>Skin contact</b> | May be absorbed through the skin in harmful amounts. Harmful in contact with skin. (based on components).       |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. Harmful if swallowed. (based on components).  |
| <b>Symptoms</b>     | Coughing and/ or wheezing.                                                                                      |

**Acute toxicity****Numerical measures of toxicity**

0 % of the mixture consists of ingredient(s) of unknown acute oral toxicity  
 0 % of the mixture consists of ingredient(s) of unknown acute dermal toxicity  
 28 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (gas)  
 13 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (vapor)  
 3 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (dust/mist)

The following values are calculated based on chapter 3.1 of the GHS document

|                                      |                |
|--------------------------------------|----------------|
| <b>ATEmix (oral)</b>                 | 665.7484 mg/kg |
| <b>ATEmix (dermal)</b>               | 1,997.00 mg/kg |
| <b>ATEmix (inhalation-dust/mist)</b> | 3.34 mg/l      |
| <b>ATEmix (inhalation-vapour)</b>    | 241.80 mg/l    |

**Component Information**

| Chemical name | Oral LD50            | Dermal LD50                                          | Inhalation LC50                                    |
|---------------|----------------------|------------------------------------------------------|----------------------------------------------------|
| Methanol      | = 6200 mg/kg ( Rat ) | = 15840 mg/kg ( Rabbit )<br>= 15800 mg/kg ( Rabbit ) | = 22500 ppm ( Rat ) 8 h<br>= 64000 ppm ( Rat ) 4 h |
| Acetone       | = 5800 mg/kg ( Rat ) | > 15700 mg/kg ( Rabbit )                             | = 50100 mg/m <sup>3</sup> ( Rat ) 8 h              |

**Delayed and immediate effects and also chronic effects from short and long term exposure**

|                                          |                                                                                                                                                                                                                                                                 |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | No information available.                                                                                                                                                                                                                                       |
| <b>Serious eye damage/irritation</b>     | No information available.                                                                                                                                                                                                                                       |
| <b>Respiratory or skin sensitization</b> | No information available.                                                                                                                                                                                                                                       |
| <b>Germ cell mutagenicity</b>            | No information available.                                                                                                                                                                                                                                       |
| <b>Carcinogenicity</b>                   | No information available.                                                                                                                                                                                                                                       |
| <b>Reproductive toxicity</b>             | No information available.                                                                                                                                                                                                                                       |
| <b>STOT - single exposure</b>            | Based on the classification criteria of the Globally Harmonized System as adopted in the country or region with which this safety data sheet complies, this product has been determined to cause systemic target organ toxicity from acute exposure. (STOT SE). |

Causes damage to organs if swallowed. Causes damage to organs in contact with skin.

**STOT - repeated exposure** No information available.

**Target organ effects** Respiratory system, Eyes, Skin, Central nervous system, Gastrointestinal tract (GI).

**Aspiration hazard** No information available.

## 12. ECOLOGICAL INFORMATION

### Toxicity

0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment

### Ecotoxicity

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                         | Crustacea                                                                                                    |
|---------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Methanol      | -                    | LC50: 13500 - 17600mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: 18 - 20mL/L (96h, <i>Oncorhynchus mykiss</i> )<br>LC50: 19500 - 20700mg/L (96h, <i>Oncorhynchus mykiss</i> )<br>LC50: =28200mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: >100mg/L (96h, <i>Pimephales promelas</i> ) | -                                                                                                            |
| Acetone       | -                    | LC50: 4.74 - 6.33mL/L (96h, <i>Oncorhynchus mykiss</i> )<br>LC50: 6210 - 8120mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: =8300mg/L (96h, <i>Lepomis macrochirus</i> )                                                                                                                   | EC50: 10294 - 17704mg/L (48h, <i>Daphnia magna</i> )<br>EC50: 12600 - 12700mg/L (48h, <i>Daphnia magna</i> ) |

### Persistence and degradability

No information available.

### Bioaccumulative potential

There is no data for this product.

### Mobility

**Mobility in soil** No information available.

**Mobility** No information available.

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Methanol      | -0.77                 |
| Acetone       | -0.24                 |

### Other adverse effects

No information available.

## 13. DISPOSAL CONSIDERATIONS

### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

## 14. TRANSPORT INFORMATION

**IMDG**

|                                                                    |                                                                           |
|--------------------------------------------------------------------|---------------------------------------------------------------------------|
| UN number or ID number                                             | UN1987                                                                    |
| UN proper shipping name                                            | ALCOHOLS, N.O.S.                                                          |
| Description                                                        | UN1987, ALCOHOLS, N.O.S. (Methanol Solution, Acetone), 3, II, (16°C C.C.) |
| Transport hazard class(es)                                         | 3                                                                         |
| Packing group                                                      | II                                                                        |
| Marine pollutant                                                   | NP                                                                        |
| Special Provisions                                                 | 274                                                                       |
| EmS-No                                                             | F-E, S-D                                                                  |
| Transport in bulk according to Annex II of MARPOL and the IBC Code | No information available                                                  |

**IATA**

|                            |                                                              |
|----------------------------|--------------------------------------------------------------|
| UN number or ID number     | UN1987                                                       |
| UN proper shipping name    | Alcohols, n.o.s.                                             |
| Description                | UN1987, Alcohols, n.o.s. (Methanol Solution, Acetone), 3, II |
| Transport hazard class(es) | 3                                                            |
| Packing group              | II                                                           |
| Special Provisions         | A3, A180                                                     |
| ERG Code                   | 3L                                                           |

**RID**

|                            |                                                              |
|----------------------------|--------------------------------------------------------------|
| UN number                  | UN1987                                                       |
| UN proper shipping name    | ALCOHOLS, N.O.S.                                             |
| Description                | UN1987, ALCOHOLS, N.O.S. (Methanol Solution, Acetone), 3, II |
| Transport hazard class(es) | 3                                                            |
| Labels                     | 3                                                            |
| Packing group              | II                                                           |
| Classification code        | F1                                                           |
| Special Provisions         | 274, 601, 640C                                               |

**ADR**

|                            |                                                            |
|----------------------------|------------------------------------------------------------|
| UN number or ID number     | 1987                                                       |
| UN proper shipping name    | ALCOHOLS, N.O.S.                                           |
| Description                | 1987, ALCOHOLS, N.O.S. (Methanol Solution, Acetone), 3, II |
| Transport hazard class(es) | 3                                                          |
| Labels                     | 3                                                          |
| Packing group              | II                                                         |
| Classification code        | F1                                                         |
| Special Provisions         | 274, 601, 640C                                             |

**ADN**

|                            |                                                              |
|----------------------------|--------------------------------------------------------------|
| UN number                  | UN1987                                                       |
| UN proper shipping name    | ALCOHOLS, N.O.S.                                             |
| Description                | UN1987, ALCOHOLS, N.O.S. (Methanol Solution, Acetone), 3, II |
| Transport hazard class(es) | 3                                                            |
| Hazard label(s)            | 3                                                            |
| Packing group              | II                                                           |
| Classification code        | F1                                                           |
| Special Provisions         | 640C, 601, 274                                               |
| Limited quantity (LQ)      | 1 L                                                          |
| Ventilation                | VE01                                                         |

Equipment Requirements PP, EX, A

Special precautions for user Special provisions from the regulations relative to the specified mode of transport are noted by numeric code. Refer to the regulations for the full text of special provisions.

## 15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**16. OTHER INFORMATION**

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 25-Aug-2021

**Revision Note** SDS sections updated. 1. 2.

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**

Revision date 25-Aug-2021

Revision Number 1.1

## 1. IDENTIFICATION

### Product identifier

**Product Name** Stop Solution

### Other means of identification

**Catalogue Number(s)** 220SM

**UN/ID no** UN3264

**Registration Number(s)** No information available

### Recommended use of the chemical and restrictions on use

**Recommended use** In-vitro laboratory reagent or component

### Supplier's details

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Diagnostic Group  
4000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pvt. Ltd.  
Bio-Rad House  
86-87, Udyog Vihar Phase IV Gurgaon  
122005  
Haryana India

Bio-Rad Laboratories (Pty) Ltd.  
34 Bolton Road  
Parkwood, Johannesburg 2193  
South Africa

### **Technical Service**

India: 91-124-4029300 or 1-800-180-1224  
South Africa: 27-11-442-85-08  
India: support.india@bio-rad.com  
South Africa: cdg\_techsupport\_eemea@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC India: 000-800-100-7141  
CHEMTREC South Africa: 0-800-983-611

## 2. HAZARDS IDENTIFICATION

### Classification of the substance or mixture

Corrosive to metals

Category 1

### GHS Label elements, including precautionary statements



**Signal word****Warning**

May be corrosive to metals

**Precautionary Statements - Prevention**

Keep only in original packaging

**Spill**

Absorb spillage to prevent material damage

**Precautionary Statements - Storage**

Store in corrosion resistant container with a resistant inner liner

**Other hazards which do not result in classification**

No information available

**3. COMPOSITION/INFORMATION ON INGREDIENTS****Substance**

Not applicable

**Mixture**

| Chemical name                  | CAS No    | Weight-% |
|--------------------------------|-----------|----------|
| Sulfuric acid<br>7664-93-9     | 7664-93-9 | 1.5      |
| Hydrochloric acid<br>7647-01-0 | 7647-01-0 | 1.5      |

**4. FIRST AID MEASURES****Description of necessary first aid measures****General advice**

Show this safety data sheet to the doctor in attendance.

**Inhalation**

Remove to fresh air. Get medical attention immediately if symptoms occur.

**Skin contact**

Wash off immediately with soap and plenty of water for at least 15 minutes. Get medical attention if irritation develops and persists.

**Eye contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Keep eye wide open while rinsing. Do not rub affected area. Get medical attention if irritation develops and persists.

**Ingestion**

Rinse mouth. Never give anything by mouth to an unconscious person. Do NOT induce vomiting. Call a physician.

**For emergency responders****Self-protection of the first aider**

Avoid contact with skin, eyes or clothing. Wear personal protective clothing (see section 8).

**Most important symptoms and effects, both acute and delayed****Symptoms**

No information available.

**Indication of immediate medical attention and special treatment needed, if necessary****Note to physicians**

Treat symptomatically.

**5. FIREFIGHTING MEASURES****Suitable Extinguishing Media****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the

surrounding environment.

**Large Fire**

CAUTION: Use of water spray when fighting fire may be inefficient.

**Unsuitable extinguishing media**

Do not scatter spilled material with high pressure water streams.

**Specific hazards arising from the chemical****Specific hazards arising from the chemical**

No information available.

**Special protective actions for fire-fighters****Special protective equipment and precautions for fire-fighters**

Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## 6. ACCIDENTAL RELEASE MEASURES

**Personal precautions, protective equipment and emergency procedures****Personal precautions**

Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required.

**Environmental precautions****Environmental precautions**

Prevent further leakage or spillage if safe to do so.

**Methods and material for containment and cleaning up****Methods for containment**

Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up**

Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations.

**Other information**

Refer to protective measures listed in Sections 7 and 8.

## 7. HANDLING AND STORAGE

**Precautions for safe handling****Advice on safe handling**

Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing.

**Conditions for safe storage, including any incompatibilities****Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from moisture. Store locked up. Keep out of the reach of children. Store away from other materials.

**Incompatible materials**

Oxidizing agent.

## 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

**Control parameters****Exposure guidelines**

| Chemical name                  | ACGIH TLV                                              | OSHA PEL                                                                             | Ontario                    | European Union                                         |
|--------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------|--------------------------------------------------------|
| Sulfuric acid<br>7664-93-9     | TWA: 0.2 mg/m <sup>3</sup> thoracic particulate matter | TWA: 1 mg/m <sup>3</sup><br>(vacated) TWA: 1 mg/m <sup>3</sup>                       | TWA: 0.2 mg/m <sup>3</sup> | TWA: 0.05 mg/m <sup>3</sup>                            |
| Hydrochloric acid<br>7647-01-0 | Ceiling: 2 ppm                                         | (vacated) Ceiling: 5 ppm<br>(vacated) Ceiling: 7 mg/m <sup>3</sup><br>Ceiling: 5 ppm | CEV: 2 ppm                 | TWA: 5 ppm<br>TWA: 8 mg/m <sup>3</sup><br>STEL: 10 ppm |



|                                |                                                                  | Ceiling: 7 mg/m <sup>3</sup>                     |                                                           | STEL: 15 mg/m <sup>3</sup>                      |                                                       |
|--------------------------------|------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------|-------------------------------------------------------|
| Chemical name                  | China                                                            | Japan Society of Occupational Health             | OEL                                                       | Australia                                       | Taiwan                                                |
| Sulfuric acid<br>7664-93-9     | TWA: 1 mg/m <sup>3</sup><br>STEL: 2 mg/m <sup>3</sup><br>G1(C**) | Ceiling: 1 mg/m <sup>3</sup>                     | TWA: 0.2 mg/m <sup>3</sup><br>STEL: 0.6 mg/m <sup>3</sup> | 1 mg/m <sup>3</sup><br>3 mg/m <sup>3</sup> STEL | TWA: 1 mg/m <sup>3</sup><br>STEL: 2 mg/m <sup>3</sup> |
| Hydrochloric acid<br>7647-01-0 | Ceiling: 7.5 mg/m <sup>3</sup><br>Ceiling                        | Ceiling: 2 ppm<br>Ceiling: 3.0 mg/m <sup>3</sup> | TWA: 1 ppm<br>STEL: 2 ppm                                 | 5 ppm Peak<br>7.5 mg/m <sup>3</sup> Peak        | Ceiling: 5 ppm<br>Ceiling: 7.5 mg/m <sup>3</sup>      |

**Appropriate engineering controls**

**Engineering controls**                      Showers  
                                                      Eyewash stations  
                                                      Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection**                      Wear safety glasses with side shields (or goggles).

**Skin and body protection**                      Wear suitable protective clothing.

**Hand protection**                              Wear suitable gloves.

**Respiratory protection**                      No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations**                      Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product. Regular cleaning of equipment, work area and clothing is recommended.

## 9. PHYSICAL AND CHEMICAL PROPERTIES

**Information on basic physical and chemical properties**

|                                              |                   |                          |                          |
|----------------------------------------------|-------------------|--------------------------|--------------------------|
| <b>Physical state</b>                        | Liquid            | <b>Odour</b>             | Odourless                |
| <b>Appearance</b>                            | clear liquid      | <b>Odour threshold</b>   | No information available |
| <b>Colour</b>                                | colourless        |                          |                          |
| <b>Property</b>                              | <b>Values</b>     | <b>Remarks • Method</b>  |                          |
| pH                                           | 3                 |                          |                          |
| Melting point / freezing point               | 0 °C / 32 °F      |                          |                          |
| Boiling point / boiling range                | 100 °C / 212 °F   |                          |                          |
| Flash point                                  |                   | No information available |                          |
| Evaporation rate                             |                   | No information available |                          |
| Flammability (solid, gas)                    |                   | No information available |                          |
| Upper/lower flammability or explosive limits |                   |                          |                          |
| Upper flammability or explosive limits       | Not applicable    |                          |                          |
| Lower flammability or explosive limits       | Not applicable    |                          |                          |
| Vapour pressure                              |                   | No information available |                          |
| Vapour density                               |                   | No information available |                          |
| Relative density                             |                   | No information available |                          |
| Solubility(ies)                              |                   |                          |                          |
| Water solubility                             | Miscible in water |                          |                          |
| Solubility in other solvents                 |                   | No information available |                          |
| Partition coefficient                        |                   | No information available |                          |
| Autoignition temperature                     |                   | No information available |                          |
| Decomposition temperature                    |                   | No information available |                          |
| Viscosity                                    |                   |                          |                          |
| Kinematic viscosity                          |                   | No information available |                          |
| Dynamic viscosity                            |                   |                          |                          |

**Other information**

**Oxidising properties**                      Not applicable

## 10. STABILITY AND REACTIVITY

### Reactivity

**Reactivity** No information available.

### Chemical stability

**Stability** Stable under normal conditions.

### Explosion data

**Sensitivity to mechanical impact** None

**Sensitivity to static discharge** None.

### Possibility of hazardous reactions

**Possibility of hazardous reactions** None under normal processing.

### Conditions to avoid

**Conditions to avoid** Exposure to air or moisture over prolonged periods.

### Incompatible materials

**Incompatible materials** Oxidizing agent.

### Hazardous decomposition products

**Hazardous decomposition products** None known based on information supplied.

## 11. TOXICOLOGICAL INFORMATION

### Information on the likely routes of exposure

#### Product Information

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available.

**Symptoms** No information available.

### Acute toxicity

#### Numerical measures of toxicity

0 % of the mixture consists of ingredient(s) of unknown acute oral toxicity

1.5 % of the mixture consists of ingredient(s) of unknown acute dermal toxicity

1.5 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (gas)

1.5 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (vapor)

0 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (dust/mist)

The following values are calculated based on chapter 3.1 of the GHS document

**ATEmix (oral)** 15,866.6667 mg/kg

**ATEmix (dermal)** 96,600.00 mg/kg

**ATEmix (inhalation-dust/mist)** 33.40 mg/l

#### Component Information

| Chemical name | Oral LD50 | Dermal LD50 | Inhalation LC50 |
|---------------|-----------|-------------|-----------------|
|---------------|-----------|-------------|-----------------|

|                   |                         |                         |                                        |
|-------------------|-------------------------|-------------------------|----------------------------------------|
| Sulfuric acid     | = 2140 mg/kg ( Rat )    | -                       | 85 - 103 mg/m <sup>3</sup> ( Rat ) 1 h |
| Hydrochloric acid | 238 - 277 mg/kg ( Rat ) | > 5010 mg/kg ( Rabbit ) | = 1.68 mg/L ( Rat ) 1 h                |

**Delayed and immediate effects and also chronic effects from short and long term exposure**

**Skin corrosion/irritation** No information available.

**Serious eye damage/irritation** No information available.

**Respiratory or skin sensitization** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

| Chemical name     | IARC    |
|-------------------|---------|
| Sulfuric acid     | Group 1 |
| Hydrochloric acid | Group 3 |

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Target organ effects** Respiratory system, Eyes, Skin, Teeth.

**Aspiration hazard** No information available.

## 12. ECOLOGICAL INFORMATION

**Toxicity**

0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment

**Ecotoxicity**

| Chemical name     | Algae/aquatic plants | Fish                                            | Crustacea                                  |
|-------------------|----------------------|-------------------------------------------------|--------------------------------------------|
| Sulfuric acid     | -                    | LC50: >500mg/L (96h, <i>Brachydanio rerio</i> ) | EC50: =29mg/L (24h, <i>Daphnia magna</i> ) |
| Hydrochloric acid | -                    | LC50: =282mg/L (96h, <i>Gambusia affinis</i> )  | -                                          |

**Persistence and degradability**

No information available.

**Bioaccumulative potential**

No information available.

**Mobility**

**Mobility in soil** No information available.

**Mobility** No information available.

**Other adverse effects**

No information available.

### 13. DISPOSAL CONSIDERATIONS

#### Disposal methods

|                                     |                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Waste from residues/unused products | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| Contaminated packaging              | Do not reuse empty containers.                                                                                  |

### 14. TRANSPORT INFORMATION

#### IMDG

|                                                                    |                                                                                                |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| UN number or ID number                                             | UN3264                                                                                         |
| UN proper shipping name                                            | CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S.                                                    |
| Description                                                        | UN3264, CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S. (Sulfuric acid, Hydrochloric acid), 8, III |
| Transport hazard class(es)                                         | 8                                                                                              |
| Packing group                                                      | III                                                                                            |
| Marine pollutant                                                   | NP                                                                                             |
| Special Provisions                                                 | 223, 274                                                                                       |
| EmS-No                                                             | F-A, S-B                                                                                       |
| Transport in bulk according to Annex II of MARPOL and the IBC Code | No information available                                                                       |

#### IATA

|                            |                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------|
| UN number or ID number     | UN3264                                                                                         |
| UN proper shipping name    | Corrosive liquid, acidic, inorganic, n.o.s.                                                    |
| Description                | UN3264, Corrosive liquid, acidic, inorganic, n.o.s. (Sulfuric acid, Hydrochloric acid), 8, III |
| Transport hazard class(es) | 8                                                                                              |
| Packing group              | III                                                                                            |
| Special Provisions         | A3, A803                                                                                       |
| ERG Code                   | 8L                                                                                             |

#### RID

|                            |                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------|
| UN number                  | UN3264                                                                                         |
| UN proper shipping name    | CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S.                                                    |
| Description                | UN3264, CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S. (Sulfuric acid, Hydrochloric acid), 8, III |
| Transport hazard class(es) | 8                                                                                              |
| Labels                     | 8                                                                                              |
| Packing group              | III                                                                                            |
| Classification code        | C1                                                                                             |
| Special Provisions         | 274                                                                                            |

#### ADR

|                            |                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------|
| UN number or ID number     | 3264                                                                                         |
| UN proper shipping name    | CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S.                                                  |
| Description                | 3264, CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S. (Sulfuric acid, Hydrochloric acid), 8, III |
| Transport hazard class(es) | 8                                                                                            |
| Labels                     | 8                                                                                            |
| Packing group              | III                                                                                          |
| Classification code        | C1                                                                                           |
| Special Provisions         | 274                                                                                          |

#### ADN

|                            |                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------|
| UN number                  | UN3264                                                                                         |
| UN proper shipping name    | CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S.                                                    |
| Description                | UN3264, CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S. (Sulfuric acid, Hydrochloric acid), 8, III |
| Transport hazard class(es) | 8                                                                                              |
| Hazard label(s)            | 8                                                                                              |
| Packing group              | III                                                                                            |

|                        |        |
|------------------------|--------|
| Classification code    | C1     |
| Special Provisions     | 274    |
| Limited quantity (LQ)  | 5 L    |
| Equipment Requirements | PP, EP |

**Special precautions for user** Special provisions from the regulations relative to the specified mode of transport are noted by numeric code. Refer to the regulations for the full text of special provisions.

## 15. REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

#### International Inventories

Contact supplier for inventory compliance status

## 16. OTHER INFORMATION

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 25-Aug-2021

**Revision Note** SDS sections updated. 1. 2.

### Key or legend to abbreviations and acronyms used in the safety data sheet

#### **Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

#### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**